



**MINNESOTA STATE UNIVERSITY, MANKATO
DEPARTMENT OF COMPUTER INFORMATION SCIENCE
COMPUTER SCIENCE PROGRAM**

DATABASE DESIGN DOCUMENT

THE MavPASS SCHEDULING PROJECT

Clients:

Dr. Laura Jacobi
MavPASS Faculty Liaison

Lina Wang
MavPASS Coordinator

THE MavPASS SCHEDULING TEAM

Madeline Ellingson
Software Engineer

Hiruy Gebregiorgis
Test Engineer

Justin Engels
Database Engineer

Marilyn Croffie
Team Lead

FACULTY COACH

Dr. Lin Chase

SUBJECT MATTER EXPERT

Dr. John Burke

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INTRODUCTION

Purpose

This document describes the design of the MavPASS Scheduling App database and maps the logical data model to the target Database Management System (DBMS), MySQL. It outlines how the data driving the app must be stored and how the data elements interrelate. This document was put together after analyzing the MavPASS Scheduling data.

Assumptions

- Entities will have their own unique tables
- The structure of entities will not change
- The relationships between entities will not change

LOGICAL DATA MODEL

	Logical Data Model	Description
Entity Names	1. Staff 2. Leaders 3. Courses 4. Sections 5. Supported Sections 6. Rooms 7. Sessions	To properly and adequately schedule MavPASS sessions, the following entities are required: Staff (teachers, coaches, etc.), leaders who hold the sessions, courses that the leader supports, sections within courses which the leader supports, supported sections which describe what sections a leader can support, rooms which sessions are held in, and the sessions themselves.
Entity Relationships	• Staff to Leaders (one to many) • Staff to Sections (one to many) • Leaders to Sessions (one to many) • Leaders to Supported Sections (one to many) • Courses to Sections (one to many) • Sections to Rooms (many to one) • Sections to Supported Sections (many to one) • Rooms to Sessions (one to many)	1. Coordinators (staff) manage all Coaches, Peer Mentors, MavPASS Leaders, and MavPASS rooms. Coaches supervise MavPASS Leaders 2. Teachers teach sessions 3. Leaders hold sessions 4. Leaders support sections 5. Courses contain multiple sections 6. Sections are held in rooms 7. Sections can be supported 8. Sessions are held in rooms

Attributes	1. Staff_ID FirstName LastName StaffType 2. Leader_ID FirstName LastName Availability PreferredAvailability NumOfSessions Coach_ID 3. Course_ID Name Abbreviation CourseNum 4. Section_ID Professor_ID Course_ID Day Time Room_ID 5. SupportedSection_ID Section_ID Leader_ID 6. Room_ID Building RoomNum RoomAvailability Capacity 7. Session_ID Day Time	1. Staff have a first and last name, and a type (teacher, coach, etc.). 2. Leaders have a first and last name, their availability and a preferable availability, the number of sessions they host, and a coach. 3. Courses have a full name alongside its abbreviation and course number. 4. Sections have a teacher/staff that teaches it, the course the section is a part of, the days and times it's held alongside the room(s) it's held in. 5. Supported sections have a leader and a section they support. 6. Rooms have the building its held in, the room number, its availability, and its capacity. 7. Sessions contain the day and time it's held at, the room it's held in, and the leader that holds it.
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	Room_ID Leader_ID	
Primary Keys	1. Staff_ID 2. Leader_ID 3. Course_ID 4. Section_ID 5. SupportedSection_ID 6. Room_ID 7. Session_ID	Each of the seven entities has its own unique id that are used as primary keys.

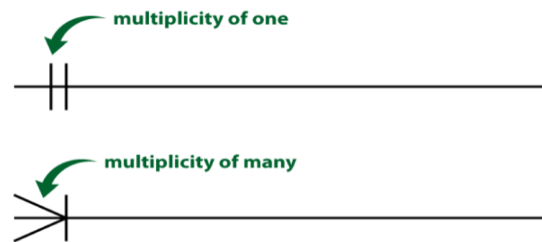
NOTATION

Entity relationships illustrate the association between two entities. They are presented as a straight line. Usually, each relationship has a name, expressed as a verb, written on the relationship line. This describes what kind of relationship connects the objects.

Cardinality

Relationships have two indicators. These are shown on both sides of the line.

- The first one (often called multiplicity) refers to the *maximum* number of times that an instance of one entity can be associated with instances in the related entity. It can be one or many.



- The second describes the *minimum* number of times one instance can be related to others. It can be zero or one, and accordingly describes the relationship as optional or mandatory.



The combination of these two indicators is always in a specific order. Placed on the outside edge of the relationship, the symbol of multiplicity comes first. The symbol indicating whether the relationship is mandatory or optional is shown after the symbol of multiplicity.

In crow's foot notation:

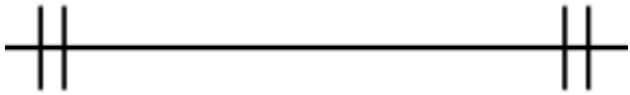
- A multiplicity of one and a mandatory relationship is represented by a straight line perpendicular to the relationship line.
- A multiplicity of many is represented by the three-pronged 'crow-foot' symbol.
- An optional relationship is represented by an empty circle.

Finally, there are four possible edges to the relationship, illustrated here:

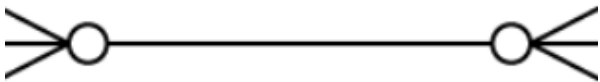
▪ zero or many	▪ one or many
▪ one and only one	▪ zero or one

Relationship degrees make them readable as :

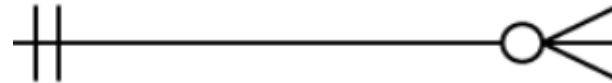
- One-to-one



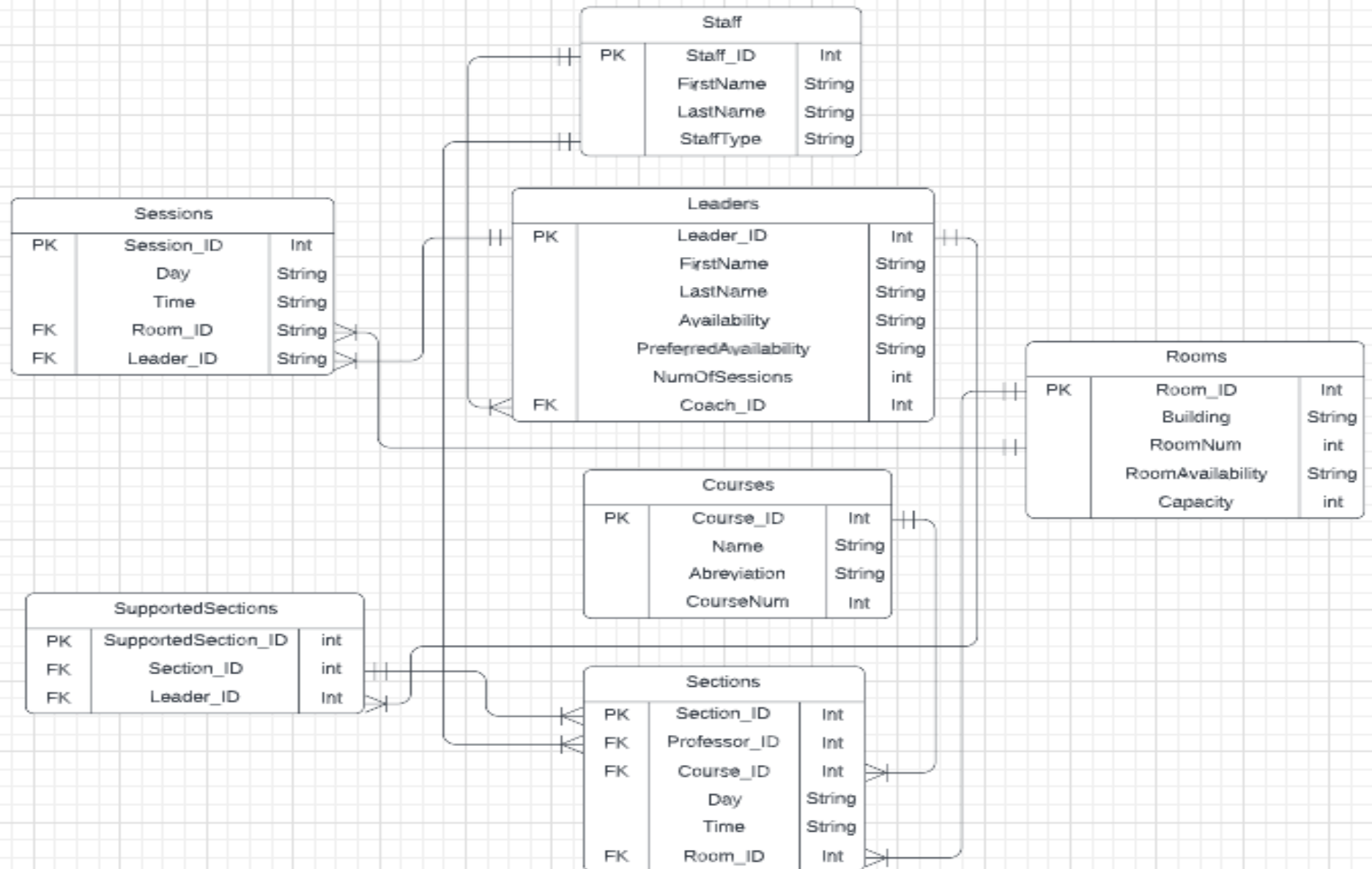
- Many-to-many



- One-to-many



ENTITY RELATIONAL DATABASE MODEL



REVISION HISTORY

Version	Date	Author(s)	Description of Change
1.0	9/27/2022	Madeline Ellingson, Marilyn Croffie, Hiruy Gebregiorgis, Justin Engels	Data Outline
2.0	10/08/2022	Madeline Ellingson, Marilyn Croffie, Hiruy Gebregiorgis, Justin Engels	Database Design Model
3.0	12/03/2022	Madeline Ellingson, Marilyn Croffie, Hiruy Gebregiorgis, Justin Engels	Final draft

APPROVALS

The undersigned acknowledge that they have reviewed the Logical Data Model proposed by The MavPASS Scheduling Team and agree with the information presented within this document. Changes to this Logical Data Model will be coordinated with and approved by, the undersigned, or their designated representatives.

Approver/Reviewer	Role	Date of Approval/Reviewal
Dr. Laura Jacobi	Client	
Lina Wang	Client	
Lin Chase	Faculty Coach	